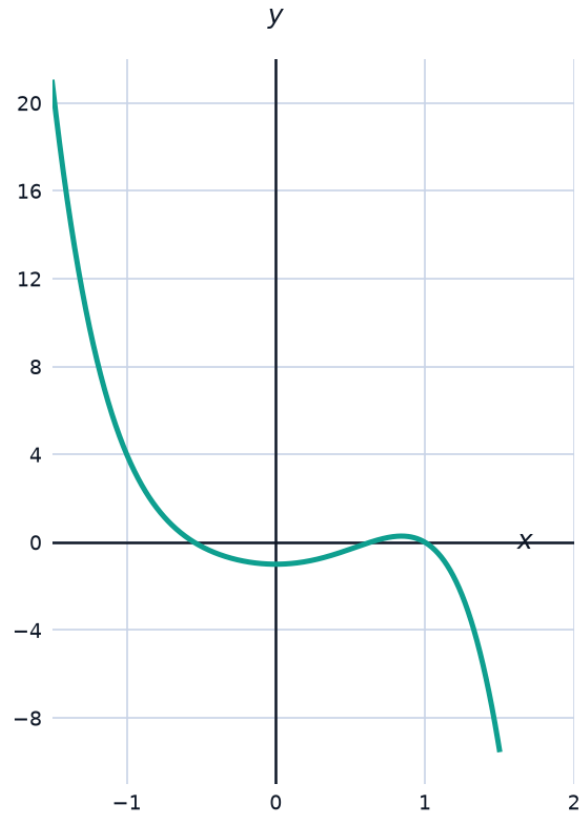


Polynomial End Behavior, Zeros & Multiplicity

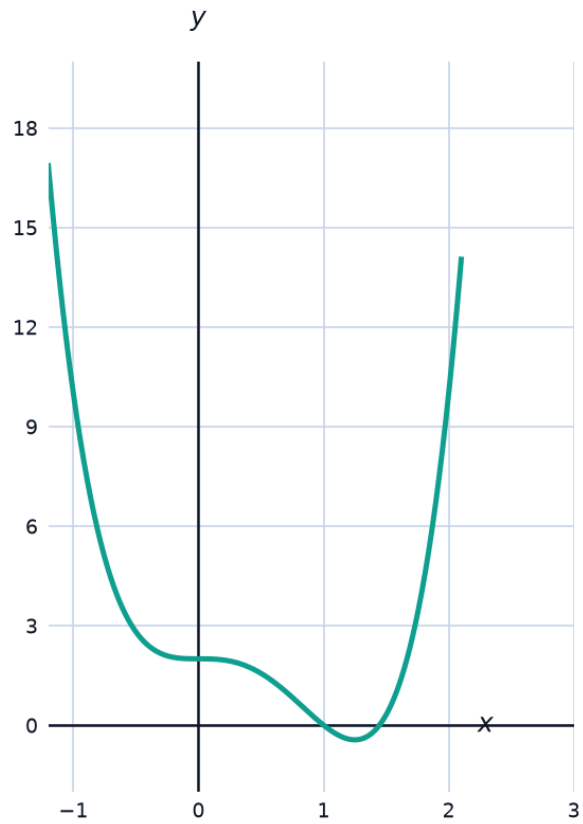


End behavior of polynomial functions

- 1 State the end behavior of $f(x) = -2x^5 + 3x^2 - 1$, using the leading term.

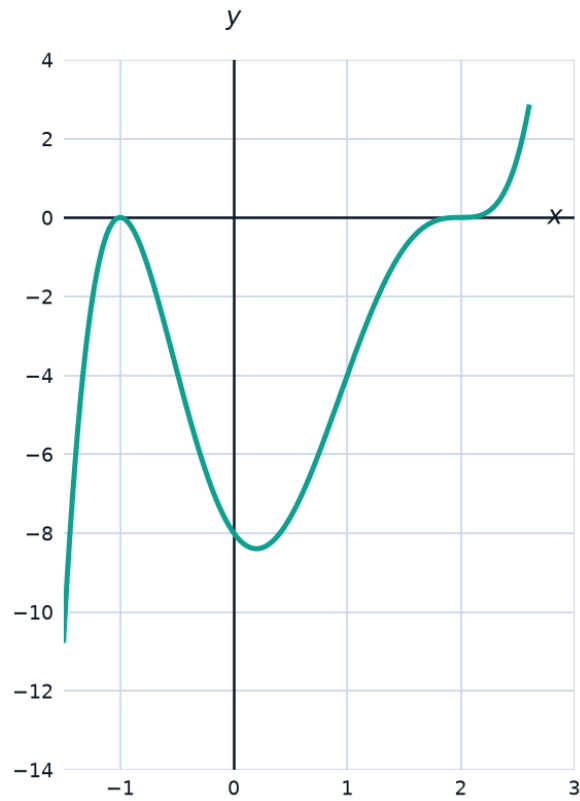


- 2 State the end behavior of $f(x) = 3x^4 - 5x^3 + 2$.

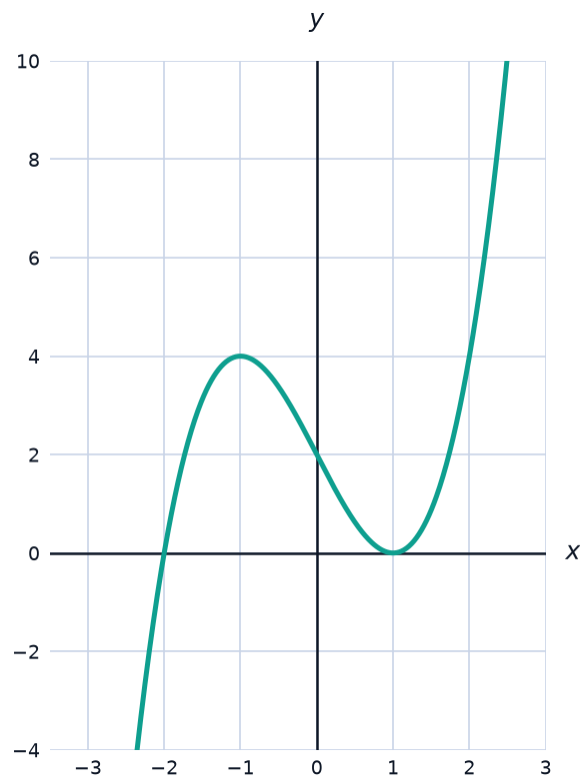


Zeros and their multiplicities

- 3 Find the zeros of $f(x) = (x - 2)^3(x + 1)^2$ and state the multiplicity and graphical behavior at each.



- 4 The graph shows $f(x) = (x + 2)(x - 1)^2$, illustrating a crossing zero at $x = -2$ and a touching zero at $x = 1$.



The Rational Zero Theorem

- 5 List all possible rational zeros of $f(x) = 2x^3 - 3x^2 - 8x + 12$ using the Rational Zero Theorem, then find the actual zeros.
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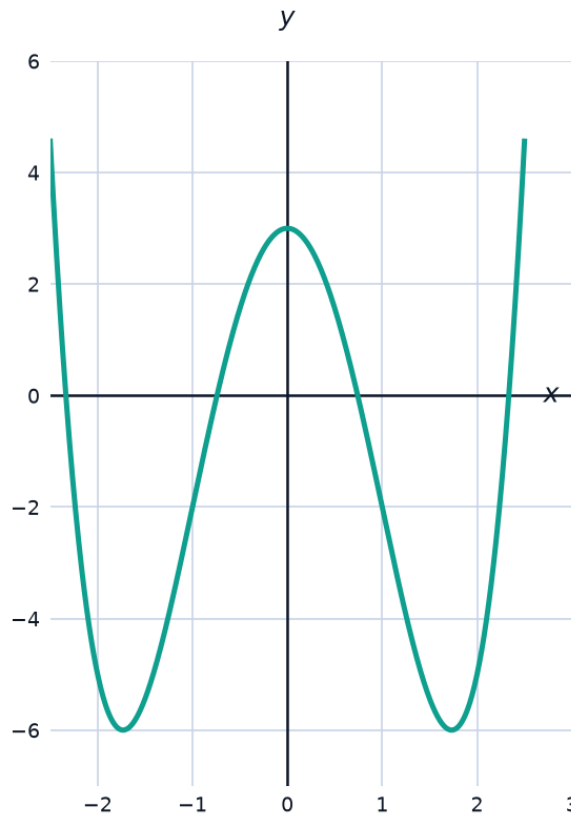
Constructing a polynomial from its zeros

- 6 Write a polynomial function of least degree with real coefficients, zeros at $x = -1$ (multiplicity 2) and $x = 3$ (multiplicity 1), and a leading coefficient of 1.
- 7 A polynomial has zeros at $x = 0$, $x = 2$, and $x = -3$, all with multiplicity 1, and passes through $(1, 8)$. Find the polynomial in the form $f(x) = a \cdot x(x - 2)(x + 3)$.
-

Local extrema and turning points

- 8 A polynomial of degree n has at most how many turning points (local extrema)? State the rule and apply it to a degree-5 polynomial.

- 9 The graph of a polynomial shows exactly 3 turning points. What is the minimum possible degree of the polynomial?



Average rate of change and concavity

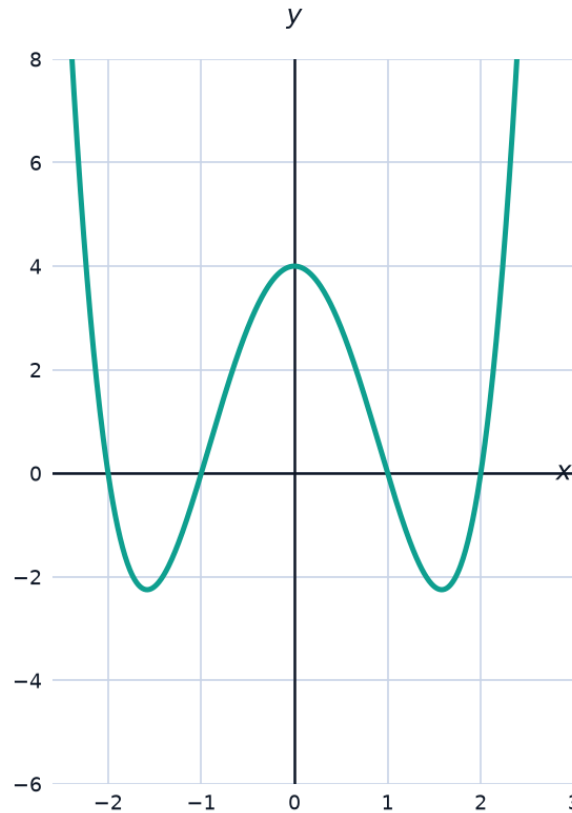
- 10 Find the average rate of change (AROC) of $f(x) = x^3 - 4x$ on $[1, 3]$.
- 11 For $f(x) = x^3 - 4x$, compare the AROC on $[0, 1]$ and $[1, 2]$, and use the comparison to determine whether f is concave up or concave down on this interval.

Sign charts and interval analysis

- 12 Build a sign chart for $f(x) = (x + 1)(x - 2)(x - 4)$ and state the intervals where $f(x) > 0$.
- 13 Use a sign chart to solve the inequality $(x - 1)^2(x + 3) \leq 0$.

Visualizing end behavior families

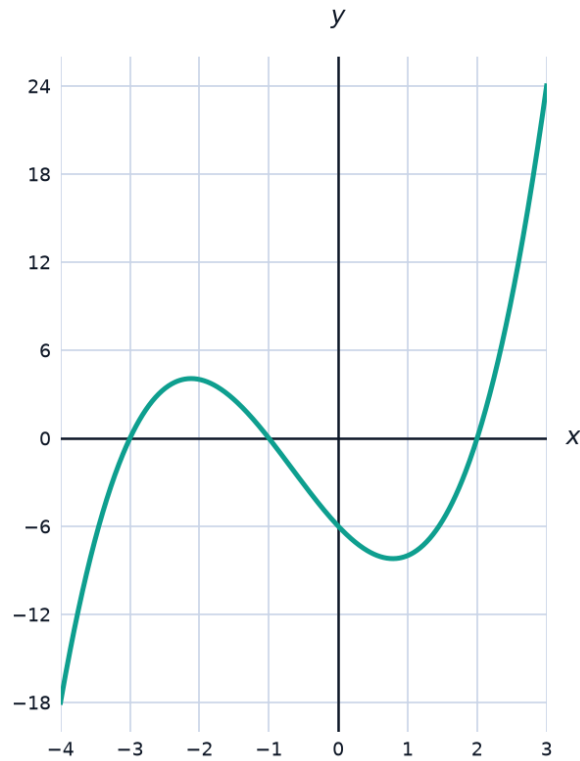
- 14 The graph shows $f(x) = x^4 - 5x^2 + 4$, a degree-4 polynomial with four real zeros.



- a State the end behavior of the graph as $x \rightarrow \pm \infty$.
- b Identify the four zeros shown and state whether each is a crossing or touching zero.
- c How many turning points does the graph display, and is this consistent with the degree?

Putting it together: a full analysis

- 15 For $f(x) = x^3 + 2x^2 - 5x - 6$: verify that $x = -1$ is a zero, factor completely, then describe the end behavior and the behavior at each zero.



Multiple choice — easy 15

- 16 State the end behavior of a polynomial with leading term x^7 .
- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
 - b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
 - c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
 - d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$

17 State the end behavior of a polynomial with leading term $9x^5$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$

18 State the end behavior of a polynomial with leading term $-5x^6$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$

19 State the end behavior of a polynomial with leading term $8x^2$.

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$

20 State the end behavior of a polynomial with leading term $6x^2$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$

21 State the end behavior of a polynomial with leading term $2x^7$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$

22 State the end behavior of a polynomial with leading term x^6 .

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$

23 State the end behavior of a polynomial with leading term $4x^3$.

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$

24 State the end behavior of a polynomial with leading term $-x^7$.

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$

25 State the end behavior of a polynomial with leading term $-2x^7$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$

26 State the end behavior of a polynomial with leading term $6x^5$.

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$

27 State the end behavior of a polynomial with leading term $4x^2$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$

28 State the end behavior of a polynomial with leading term $8x^5$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$

29 State the end behavior of a polynomial with leading term $7x^7$.

- a. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- c. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$

30 State the end behavior of a polynomial with leading term $-7x^2$.

- a. As $x \rightarrow \pm\infty$, $f(x) \rightarrow -\infty$
- b. As $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow -\infty$
- c. As $x \rightarrow \pm\infty$, $f(x) \rightarrow +\infty$
- d. As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$; as $x \rightarrow +\infty$, $f(x) \rightarrow +\infty$

Multiple choice — medium 15

31 Find the degree of $f(x) = (x - 4)^1(x + 6)^2(x + 7)^1$.

- a. 2
- b. 5
- c. 7
- d. 4

32 Find the degree of $f(x) = (x + 2)^2(x + 3)^1(x + 8)^1$.

- a. 5
- b. 2
- c. 6
- d. 4

33 Find the degree of $f(x) = (x - 9)^2(x + 2)^2(x - 7)^3$.

a. 12

b. 7

c. 8

d. 3

34 Find the degree of $f(x) = (x + 3)^2(x - 6)^1(x - 2)^2$.

a. 4

b. 6

c. 5

d. 2

35 Find the degree of $f(x) = (x + 8)^3(x + 6)^2(x - 6)^3$.

a. 8

b. 9

c. 18

d. 3

36 Find the degree of $f(x) = (x - 3)^2(x + 8)^3(x + 1)^3$.

a. 8

b. 9

c. 3

d. 18

37 Find the degree of $f(x) = (x + 9)^2(x - 3)^1(x - 6)^2$.

a. 2

b. 4

c. 6

d. 5

38 Find the degree of $f(x) = (x + 7)^2(x - 7)^1(x - 1)^1$.

a. 4

b. 5

c. 7

d. 2

39 Find the degree of $f(x) = (x + 9)^2(x + 6)^1(x - 7)^1$.

a. 4

b. 5

c. 3

d. 2

40 Find the degree of $f(x) = (x + 7)^2(x - 1)^2(x + 6)^3$.

a. 3

b. 8

c. 7

d. 12

41 Find the degree of $f(x) = (x + 6)^1(x + 1)^3(x - 8)^2$.

a. 7

b. 3

c. 6

d. 8

42 Find the degree of $f(x) = (x - 8)^2(x + 4)^2(x + 2)^2$.

a. 2

b. 8

c. 7

d. 6

43 Find the degree of $f(x) = (x - 1)^3(x + 2)^2(x - 4)^3$.

a. 8

b. 18

c. 9

d. 3

44 Find the degree of $f(x) = (x - 9)^3(x + 1)^2(x + 3)^1$.

a. 3

b. 6

c. 7

d. 8

45 Find the degree of $f(x) = (x - 7)^3(x - 2)^3(x + 7)^3$.

- a. 27
- b. 3
- c. 10
- d. 9

Multiple choice — hard 15

46 For $f(x) = (x + 2)^1(x - 8)^3$, describe the multiplicity and graphical behavior at $x = -2$.

- a. Multiplicity 1 (odd) — the graph crosses the x-axis at $x = -2$.
- b. Multiplicity 1 (even) — the graph touches and turns back without crossing the x-axis at $x = -2$.
- c. Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 8$.
- d. Multiplicity 3 (odd) — the graph crosses the x-axis at $x = -2$.

47 For $f(x) = (x - 5)^4(x - 3)^2$, describe the multiplicity and graphical behavior at $x = 5$.

- a. Multiplicity 2 (even) — the graph touches and turns back without crossing the x-axis at $x = 5$.
- b. Multiplicity 4 (odd) — the graph crosses the x-axis at $x = 5$.
- c. Multiplicity 4 (even) — the graph touches and turns back without crossing the x-axis at $x = 5$.
- d. Multiplicity 4 (even) — the graph touches and turns back without crossing the x-axis at $x = 3$.

48 For $f(x) = (x - 9)^1(x + 7)^2$, describe the multiplicity and graphical behavior at $x = 9$.

- a. Multiplicity 1 (odd) — the graph crosses the x-axis at $x = -7$.
- b. Multiplicity 1 (even) — the graph touches and turns back without crossing the x-axis at $x = 9$.
- c. Multiplicity 2 (even) — the graph touches and turns back without crossing the x-axis at $x = 9$.
- d. Multiplicity 1 (odd) — the graph crosses the x-axis at $x = 9$.

- 49 For $f(x) = (x - 8)^1(x - 7)^2$, describe the multiplicity and graphical behavior at $x = 8$.
- a. Multiplicity 1 (odd) — the graph crosses the x -axis at $x = 8$.
 - b. Multiplicity 1 (odd) — the graph crosses the x -axis at $x = 7$.
 - c. Multiplicity 2 (even) — the graph touches and turns back without crossing the x -axis at $x = 8$.
 - d. Multiplicity 1 (even) — the graph touches and turns back without crossing the x -axis at $x = 8$.
- 50 For $f(x) = (x - 5)^2(x + 9)^1$, describe the multiplicity and graphical behavior at $x = 5$.
- a. Multiplicity 2 (odd) — the graph crosses the x -axis at $x = 5$.
 - b. Multiplicity 1 (odd) — the graph crosses the x -axis at $x = 5$.
 - c. Multiplicity 2 (even) — the graph touches and turns back without crossing the x -axis at $x = -9$.
 - d. Multiplicity 2 (even) — the graph touches and turns back without crossing the x -axis at $x = 5$.
- 51 How many possible rational zeros does $f(x) = x^n + \dots + 6$ (leading coefficient 1, constant term 6) have, by the Rational Zero Theorem?
- a. 4
 - b. 10
 - c. 6
 - d. 8
- 52 How many possible rational zeros does $f(x) = x^n + \dots + 10$ (leading coefficient 1, constant term 10) have, by the Rational Zero Theorem?
- a. 10
 - b. 4
 - c. 8
 - d. 6

- 53 How many possible rational zeros does $f(x) = x^n + \dots + 25$ (leading coefficient 1, constant term 25) have, by the Rational Zero Theorem?
- a. 6
 - b. 8
 - c. 4
 - d. 3
- 54 How many possible rational zeros does $f(x) = x^n + \dots + 16$ (leading coefficient 1, constant term 16) have, by the Rational Zero Theorem?
- a. 12
 - b. 10
 - c. 5
 - d. 8
- 55 How many possible rational zeros does $f(x) = x^n + \dots + 5$ (leading coefficient 1, constant term 5) have, by the Rational Zero Theorem?
- a. 6
 - b. 2
 - c. 10
 - d. 4
- 56 A polynomial has zeros at $x = 0$, $x = 1$, $x = 2$ (each multiplicity 1) and passes through $(-1, -18)$. Find a in $f(x) = a(x)(x - 1)(x - 2)$.
- a. -3
 - b. 3
 - c. 6
 - d. 4

- 57 A polynomial has zeros at $x = -3$, $x = 6$, $x = -1$ (each multiplicity 1) and passes through $(0, -72)$. Find a in $f(x) = a(x + 3)(x - 6)(x + 1)$.
- a. 4
 - b. 8
 - c. -4
 - d. 5
- 58 A polynomial has zeros at $x = 5$, $x = 0$, $x = 4$ (each multiplicity 1) and passes through $(-6, 660)$. Find a in $f(x) = a(x - 5)(x)(x - 4)$.
- a. -1
 - b. -2
 - c. 1
 - d. 0
- 59 A polynomial has zeros at $x = -6$, $x = 0$, $x = -5$ (each multiplicity 1) and passes through $(-3, 90)$. Find a in $f(x) = a(x + 6)(x)(x + 5)$.
- a. -4
 - b. -10
 - c. 5
 - d. -5
- 60 A polynomial has zeros at $x = -2$, $x = -4$, $x = 6$ (each multiplicity 1) and passes through $(0, -96)$. Find a in $f(x) = a(x + 2)(x + 4)(x - 6)$.
- a. 4
 - b. 2
 - c. -2
 - d. 3